

# Probability Theory

## Continuous Random Variables and the Variance

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# (Absolutely) Continuous Rvs

The properties of the PMF  $p_X(x)$  of a discrete rv  $X$  :

- ①  $p_X(x) \geq 0$ ;
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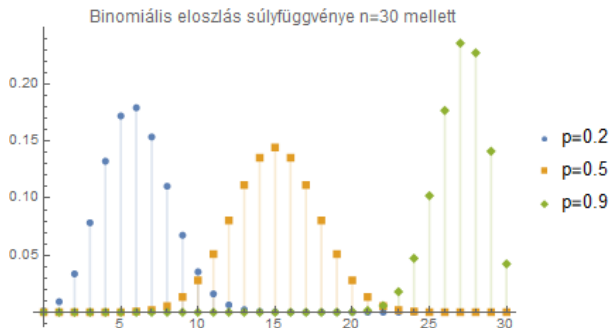
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- $\mathbb{P}(X \in B) = \int_B f_X(x) dx$ .
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Notation:  $\int_B f_X(x) dx = \int_a^b f_X(x) dx$  when  $B = (a, b)$  is an interval.

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# The Standard Normal Distribution

## Definition

*If  $Z$  is a r.v. with PDF*

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}, \quad x \in (-\infty, \infty)$$

*then it is called standard normally distributed.*

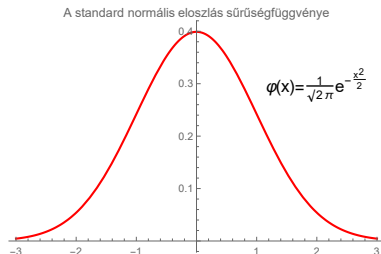
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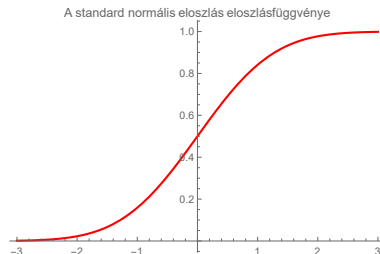
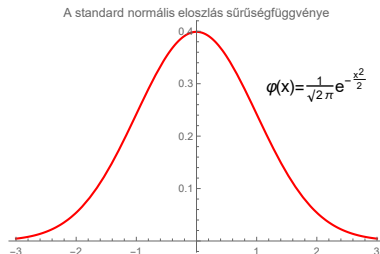
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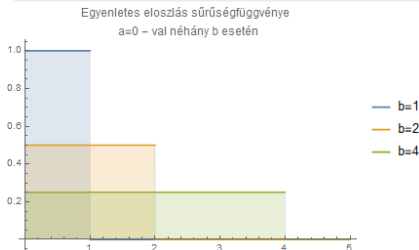
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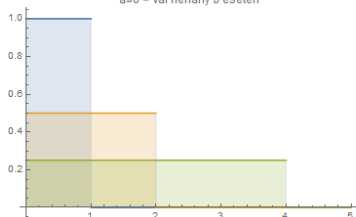
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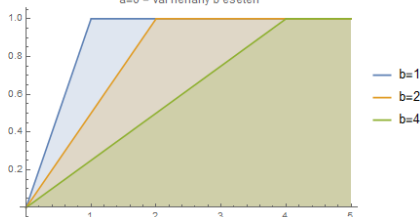
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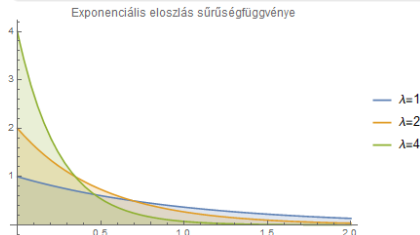
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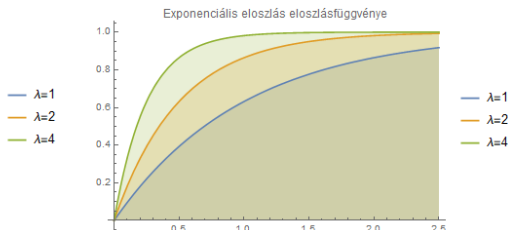
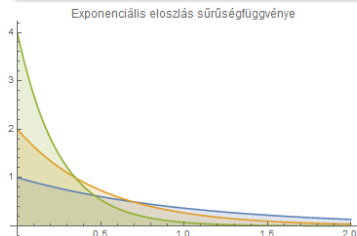
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